

Yingru Li

szrlee@gmail.com • +65 8747 5669 • richardli.xyz

Curriculum Vitae

EDUCATION

The Chinese University of Hong Kong (CUHK)

2025

Ph.D. in Computer Science and Information Engineering.

Advisor: Zhi-Quan (Tom) Luo.

Committee: Xinyun Chen, Baoxiang Wang, John C.S. Lui, Benjamin Van Roy (Stanford & DeepMind).

Huazhong University of Science and Technology (HUST)

2020

B.E./M.S. in Computer Science. Ranked 1st/134 overall; 1st/26 in Computer Theory and Software.

PROFESSIONAL EXPERIENCE

xAI - SpaceX

Singapore

Member of Technical Staff

December 2025 – Present

- Drive large-scale (“big-run”) reinforcement learning for the Grok models (Grok 4.x, Grok Build), spanning both the **science** and the **infrastructure** of RL training.
- **Science of RL** — stability, efficiency, and the core algorithms and recipes. Identified and resolved critical training instabilities that benefit every stage of the RL pipeline, directly enabling the longest stable RL run to date.
- **Long-horizon RL** — developing recipes for higher intelligence per token and recursive self-improvement: RL training with state compaction, self-distillation RL for dense credit assignment, and extracting more than one bit of information per rollout — solving long-horizon credit assignment and enabling continual learning.
- **RL infrastructure** — built and scaled the systems behind these recipes, including big-run training across $\gg 10,000$ GPUs.

ByteDance

Singapore

LLM Algorithm Research Scientist

2025 – December 2025

- Led research on RL training stability and efficiency for LLM reasoning and agents.
- Diagnosed and formalized the **RL Collapse** phenomenon, tracing instability to implicit off-policy-ness caused by the training–inference mismatch; developed Trust Region Masking, Dynamic Vocabulary Pruning, the Optimal Token Baseline, and a simple LR-scheduling fix.
- Co-led SimpleTIR (end-to-end multi-turn tool-integrated reasoning) and Dr. Kernel (Triton kernel generation agent via RL).

Tencent AI & Robotics X

Shenzhen, China

Student Research Staff Member, Agent Center

2019 – 2022

- Led HyperAgent (ICML 2024) and HyperDQN (ICLR 2022) for scalable exploration in deep RL, and Divergence-Augmented Policy Optimization (NeurIPS 2019) for stable off-policy data reuse.

Microsoft Research Lab – Asia

Beijing, China

Research Intern, Theory Center

2016

- Research on combinatorial optimization and reinforcement / bandit learning algorithms.

PUBLICATIONS

Full list at [Google Scholar](#) (> 800 citations, h-index 10). * denotes equal contribution; † denotes corresponding author.

1. [The Optimal Token Baseline: Variance Reduction for Long-Horizon LLM-RL](#)

Yingru Li, Jiawei Xu, Ziniu Li, Jiakai Liu, Wei Liu, Yuxuan Tong, Longtao Zheng, Zhenghai Xue, Yaxiang Zhang, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *ICML* 2026.

2. **Dr. Kernel: Reinforcement Learning Done Right for Triton Kernel Generations**
Wei Liu, Jiawei Xu, **Yingru Li**, Longtao Zheng, Tianjian Li, Qian Liu, Junxian He. *ICML 2026*.
3. **When Speed Kills Stability: Demystifying RL Collapse from the Training-Inference Mismatch**
Jiacai Liu*, **Yingru Li***,[†], Yuqian Fu, Jiawei Wang, Qian Liu, Yu Shen. *Technical Report, 2025*.
4. **Trust Region Masking for Long-Horizon LLM Reinforcement Learning**
Yingru Li, Jiakai Liu, Jiawei Xu, Yuxuan Tong, Ziniu Li, Qian Liu, Baoxiang Wang. *ICML 2026*.
5. **Harnessing Uncertainty: Entropy-Modulated Policy Gradients for Long-Horizon LLM Agents**
Jiawei Wang, Jiakai Liu, Yuqian Fu, **Yingru Li**, Xintao Wang, Yuan Lin, Yu Yue, Lin Zhang, Yang Wang, Ke Wang. *ICML 2026*.
6. **SimpleTIR: End-to-End Reinforcement Learning for Multi-Turn Tool-Integrated Reasoning**
Zhenghai Xue, Longtao Zheng, Qian Liu, **Yingru Li**, Xiaosen Zheng, Zejun Ma, Bo An. *ICLR 2026*.
7. **Owl: Optimized Workforce Learning for General Multi-Agent Assistance in Real-World Task Automation**
Mengkan Hu, Yuhang Zhou, Wendong Fan, Yuzhou Nie, Bowei Xia, Tao Sun, Ziyu Ye, Zhaoxuan Jin, **Yingru Li**, Qiguang Chen, Zeyu Zhang, Yifeng Wang, Qianshuo Ye, Bernard Ghanem, Ping Luo, Guohao Li. *NeurIPS 2025*.
8. **Scalable Exploration via Ensemble++**
Yingru Li, Jiawei Xu, Baoxiang Wang, Zhi-Quan Luo. *NeurIPS 2025*.
9. **Q-Star Meets Scalable Posterior Sampling: Bridging Theory and Practice via HyperAgent**
Yingru Li, Jiawei Xu, Lei Han, Zhi-Quan Luo. *ICML 2024*.
10. **Prior-Dependent Analysis of Posterior Sampling RL with Function Approximation**
Yingru Li, Zhi-Quan Luo. *AISTATS 2024*.
11. **HyperDQN: A Randomized Exploration Method for Deep Reinforcement Learning**
Ziniu Li, **Yingru Li**[†], Yushun Zhang, Tong Zhang, Zhi-Quan Luo. *ICLR 2022*.
12. **Divergence-Augmented Policy Optimization**
Qing Wang*, **Yingru Li***, Jiechao Xiong, Tong Zhang. *NeurIPS 2019*.
13. **Beyond Precision: Training-Inference Mismatch is an Optimization Problem and Simple LR Scheduling Fixes It**
Yaxiang Zhang*, **Yingru Li***,[†], Jiakai Liu, Jiawei Xu, Ziniu Li, Qian Liu, Haoyuan Li. *arXiv 2026*.
14. **Taming the Tail: Stable LLM Reinforcement Learning via Dynamic Vocabulary Pruning**
Yingru Li, Jiawei Xu, Jiakai Liu, Yuxuan Tong, Ziniu Li, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *arXiv 2025*.
15. **A Note on Hybrid Online RL and Imitation Learning for LLMs: Formulations and Algorithms**
Yingru Li, Ziniu Li, Jiakai Liu. *arXiv 2025*.
16. **Logit Dynamics in Softmax Policy Gradient Methods**
Yingru Li. *arXiv 2025*.
17. **Scaling Flaws of Verifier-Guided Search in Mathematical Reasoning**
Fei Yu, **Yingru Li**, Benyou Wang. *arXiv 2025*.
18. **Uncertainty-Aware Search and Value Models: Mitigating Search Scaling Flaws in LLMs**
Fei Yu, **Yingru Li**, Benyou Wang. *arXiv 2025*.
19. **Probability Tools for Sequential Random Projection**
Yingru Li. *arXiv 2024*.
20. **Simple, Unified Analysis of Johnson-Lindenstrauss with Applications**
Yingru Li. *arXiv 2024*.

PATENTS

- Liangqi Liu, Wenqiang Pu, **Yingru Li**, Zhi-Quan Luo. *Game-Strategy Update Method, Device, and Medium for Unknown Matrix Games*. Chinese invention patent application, CN117787419A (Published Mar. 2024).

RESEARCH IMPACT & ADOPTION

The training–inference mismatch / RL-collapse line of work (*When Speed Kills Stability*) has been adopted across major open-source RL training stacks and frontier models:

- DeepSeek-V3.2 adopts the geometric sequence masking to address the training–inference mismatch and

off-policy training instability ([arXiv:2512.02556](#)).

- Integrated into [veRL](#) (v0.6.0+): geometric-mean importance-sampling masking / rejection (Geo-MIS / Geo-RS) and [rollout correction](#) for general off-policy training; Trust Region Masking and the Optimal Token Baseline are also available in veRL.
- Integrated into [Slime](#) and [Miles](#). Discussed in a [vLLM engineering blog](#) (bitwise-consistent on-policy RL) and a [SGLang Blog](#).
- Used by [Cognition's SWE-grep](#) (sequence-level masked importance sampling) and discussed by the [Qwen team](#).

AWARDS & HONORS

- **Best Paper Award**, 3rd Doctoral and Postdoctoral Daoyuan Academic Forum, 2024.
- **Best Student Paper**, IEEE Sensor Array and Multichannel Signal Processing Workshop (SAM), 2024.
- **SRIBD Ph.D. Fellowship (Gold Class)**, Shenzhen Research Institute of Big Data, 2023.
- **Presidential Ph.D. Fellowship**, The Chinese University of Hong Kong, 2019–2023.
- **Tencent AI Ph.D. Fellowship**, Tencent & The Chinese University of Hong Kong, 2018.
- **National Scholarship (China)** — highest national academic honor, top 0.2% nationwide, 2018.
- **Qiming Star Award** — 1 of 5 recipients from 7,112 undergraduates, HUST, 2016.
- **First Prize**, Parallel Computation and Application Contest (PAC), Intel & CCF, 2015.
- **First Prize**, China National Mathematics Olympiad (Province-level Math League), 2012.

PEER REVIEW & PROFESSIONAL SERVICE

Reviewer for the leading peer-reviewed venues in machine learning and artificial intelligence. NeurIPS, ICML, and ICLR are the three top-tier conferences in the field; AISTATS is a leading venue in machine-learning theory and statistics; TMLR is a flagship ML journal.

Venue	Role	Papers reviewed
NeurIPS (2023, 2024, 2025)	Reviewer	6 + 6 + 5 = 17
ICLR (2024, 2025, 2026)	Reviewer	2 + 3 + 5 = 10
ICML (2025, 2026)	Reviewer [†]	5 + 6 = 11
AISTATS (2025)	Reviewer	2
TMLR (ongoing)	Reviewer	3
Total (2023–2026)		43 papers

- † **ICML 2026 Gold Reviewer** — recognized by the ICML Program Chairs as among the top reviewers for the conference, based on area-chair evaluations of the quality of reviews submitted; awarded complimentary registration.
- Admitted to the NeurIPS 2026 and AISTATS 2026 reviewer pools (paper assignments pending).
- **28 additional papers** reviewed across 9 affiliated workshops, including ICLR Workshop “Bridging the Gap Between Practice and Theory in Deep Learning,” ICML Workshop “Aligning RL Experimentalists and Theorists,” and NeurIPS Workshops on Bayesian Decision-making & Uncertainty and RL4RealLife.
- **Session Chair**, 2 sessions on “Integrating Generative AI with Sequential Decision-making,” INFORMS Annual Meeting 2024.
- **Organizer / Chair**, Reinforcement Learning Seminar, CUHK (8 semesters, 2019–2022).

INVITED TALKS & ORAL PRESENTATIONS

- **Revisiting Trust Region Policy Optimization in the Era of LLMs** — Invited talk, QingkeAI (Dec. 2025).
- **LLM-RL Instability and the Training–Inference Mismatch** — Invited talks at SLAI and the Bay Area RL community (Nov. 2025).

- **Tackling Data Scarcity for Trustworthy Agents** — Invited talk, ETH Zurich (Nov. 2024).
- **Exploration at Scale: Theory, Algorithms & Applications** (*a.k.a. Scalable Uncertainty Quantification for Exploration and LLM Reasoning*) — Invited talks at MIT (Jul. 2024), Princeton University (May 2024), INFORMS Annual Meeting (Seattle, Oct. 2024), International Symposium on Mathematical Programming / ISMP (Montréal, Jul. 2024), INFORMS Optimization Society Conference at Rice University (Mar. 2024), and RLChina (Jun. 2024); contributed talk at the 3rd Daoyuan Academic Forum (Jan. 2024).
- **No-Regret Learning in Unknown Games with Applications** — Invited talk, RL Theory Student Workshop, Nanjing University (Aug. 2022); contributed talk, 2nd Daoyuan Academic Forum (Aug. 2022).
- **HyperDQN: A Randomized Exploration Method for Deep RL** — Contributed talk, NeurIPS Workshop on Ecological Theory of RL (Dec. 2021).

SELECTED MEDIA COVERAGE

- Featured for the Qiming Star Award in the HUST campus newspaper and HUST Online, 2016.

TEACHING

Teaching Assistant, The Chinese University of Hong Kong (Shenzhen) — duties delivered in English (tutorials, grading, laboratory sessions): Stochastic Processes (Fall 2018), Optimization II (Spring 2019), Distributed and Parallel Computing (Fall 2019), Reinforcement Learning (Fall 2020), Matrix Analysis (Spring 2021), and Deep Learning and Their Applications (Spring 2022).